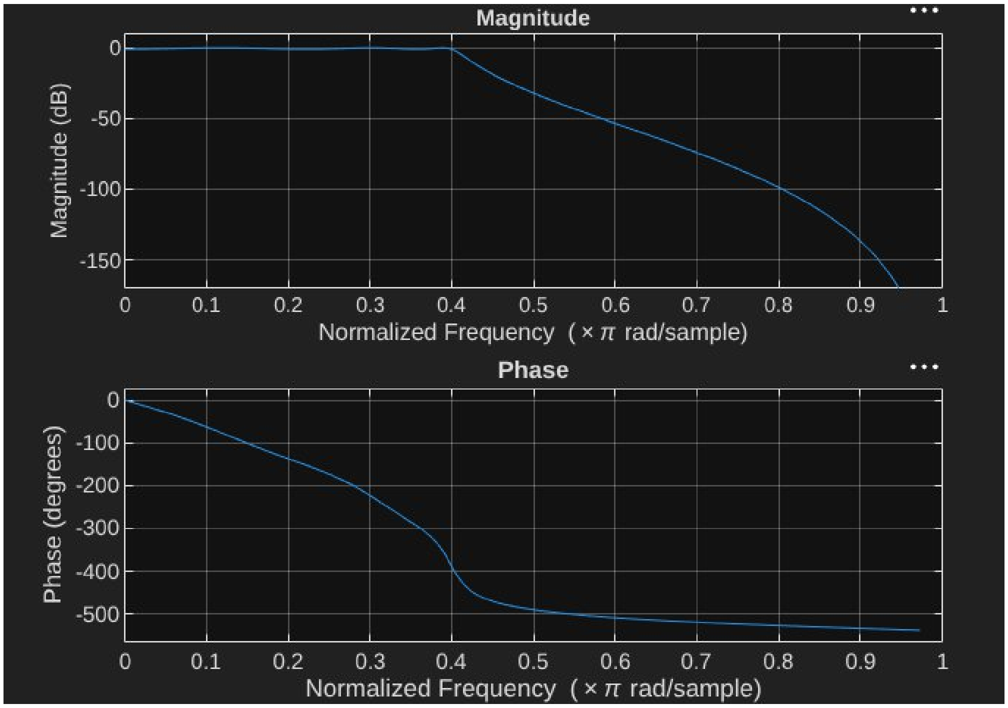
2.1. Design and Performance Evaluation of a Chebyshev Type-I IIR Low-Pass Filter using MATLAB

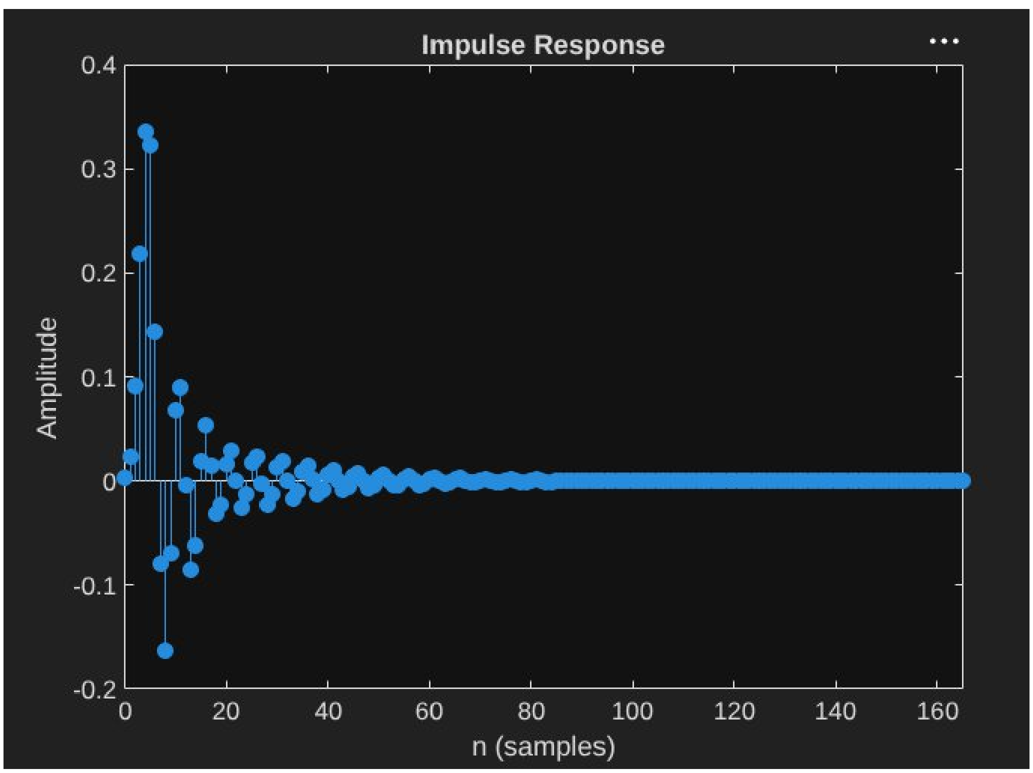
Program (MATLAB Code)

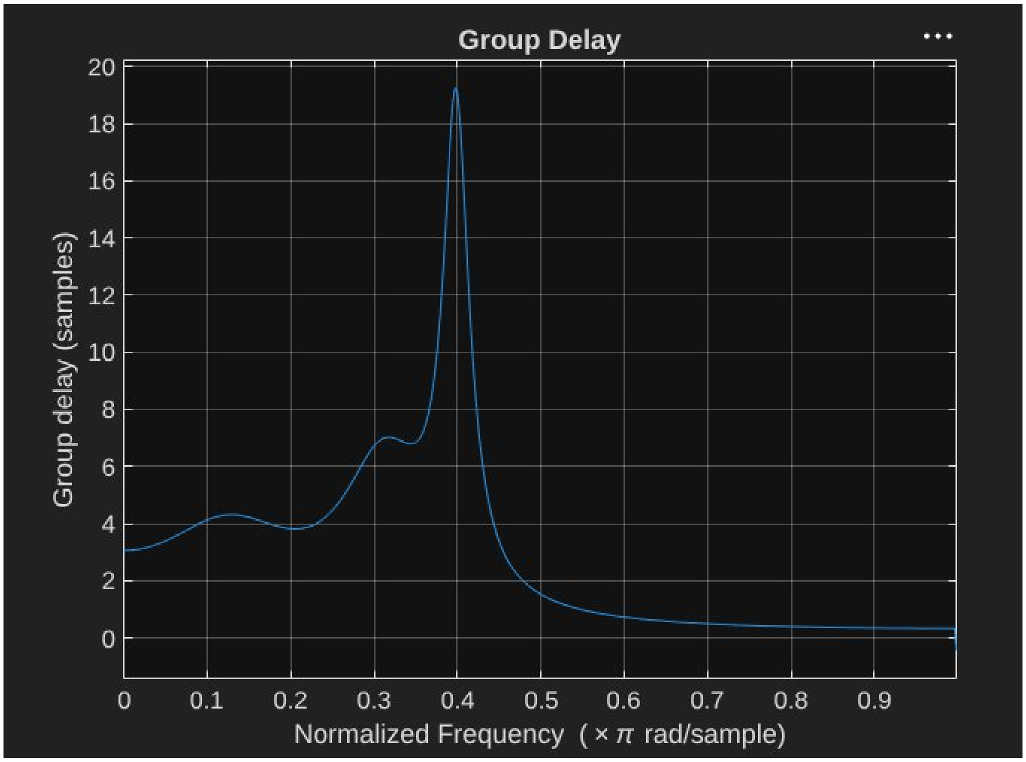
```
clc; clear;

Wp=0.4; Ws=0.55; Rp=1; As=40;

[N,Wn]=cheb1ord(Wp,Ws,Rp,As);

[b,a]=cheby1(N,Rp,Wn);

fprintf('Filter Order = %d\n',N);

fprintf('Cutoff Frequency = %.2f * pi rad/sample\n',Wn);

if all(abs(roots(a))<1)

disp('System is STABLE');

else

disp('System is UNSTABLE');

end

% -------- GRAPHS --------

figure; freqz(b,a); % Magnitude + Phase

figure; impz(b,a); % Impulse Response

figure; grpdelay(b,a); % Group Delay

figure; zplane(b,a); % Pole-Zero
```

Objective 1:

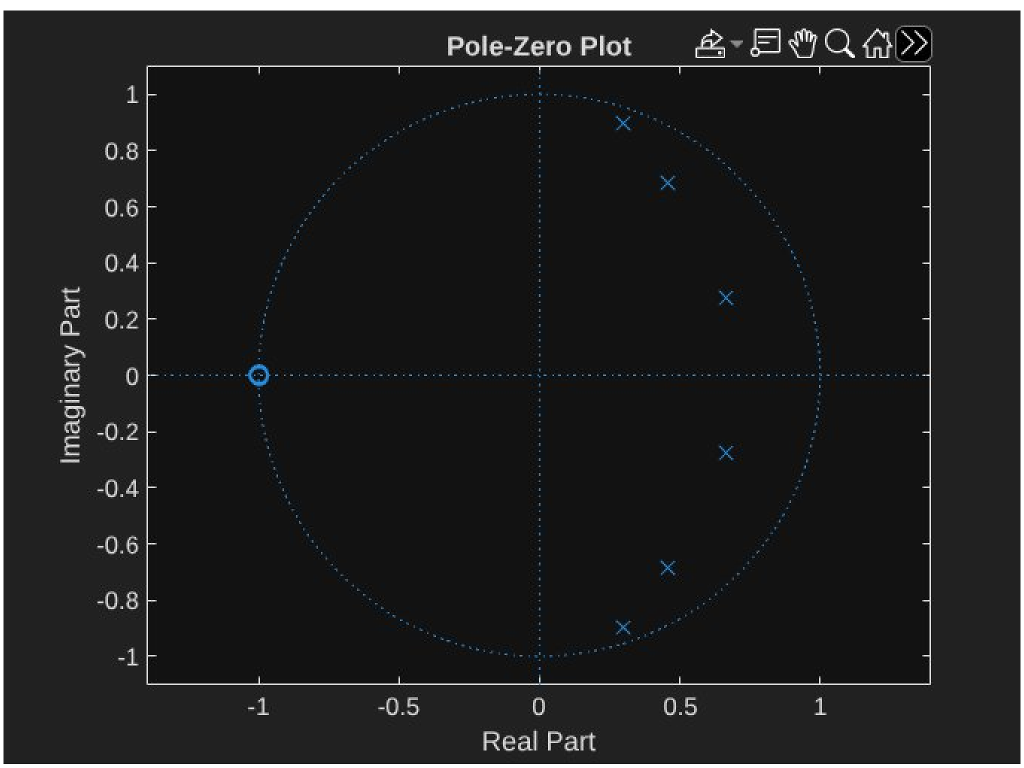


Objective 2:




Group Delay
Group delay (samples)
20
18
16
14
12
10
8
6
4
2
0
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
Normalized Frequency (× π rad/sample)

Ojective 3:


Pole-Zero Plot
Imaginary Part
1
0.8
0.6
0.4
0.2
0
-0.2
-0.4
-0.6
-0.8
-1
-1
-0.5
0
0.5
1
Real Part

2.2. Design and Performance Evaluation of a Chebyshev Type-I IIR High-PassFilter

Program (MATLAB Code)

```
clc;

clear;

close all;

% Specifications

Rp = 1; % Passband Ripple (dB)

Rs = 40; % Stopband Attenuation (dB)

Wp = 0.5; % Passband Edge

Ws = 0.35; % Stopband Edge

% Find Filter Order

[n,Wn] = cheb1ord(Wp,Ws,Rp,Rs);

% Design High-Pass Filter

[b,a] = cheby1(n,Rp,Wn,'high');

% Frequency Response

figure;

freqz(b,a);

title('Frequency Response');

% Impulse Response

figure;

impz(b,a);

title('Impulse Response');

% Group Delay

figure;
```

```
grpdelay(b,a);

title('Group Delay');

% Pole-Zero Plot

figure;

zplane(b,a);

title('Pole-Zero Plot');

fprintf('Filter Order = %d\n',n);

fprintf('Cutoff Frequency = %.4f\n',Wn);
```

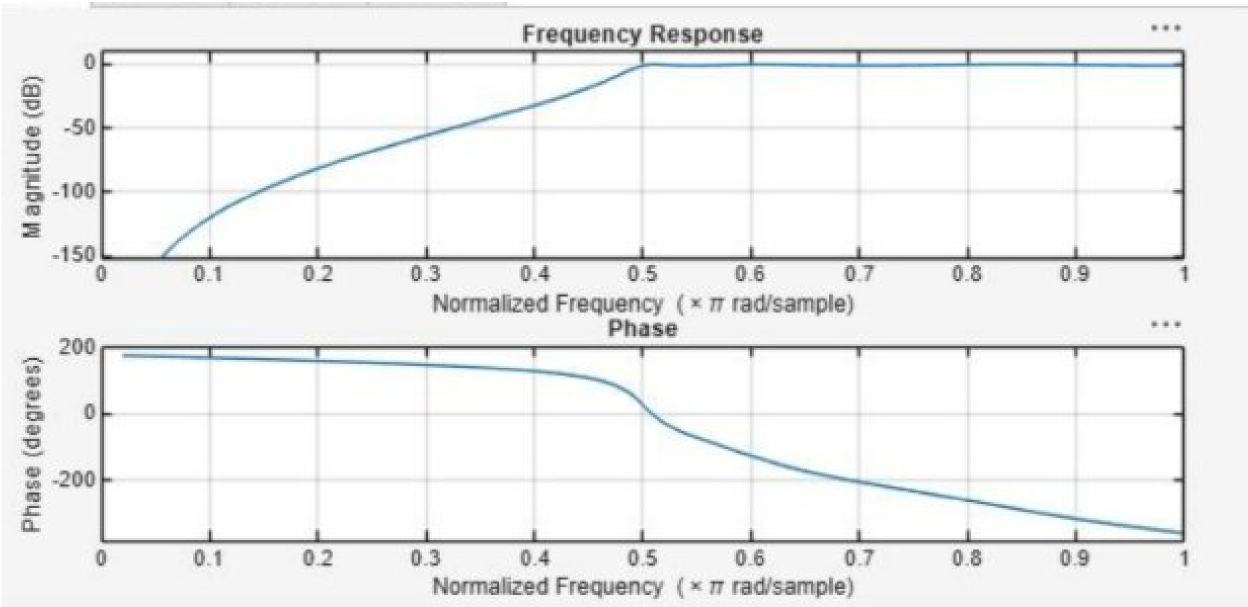
Output:

Objective-1:

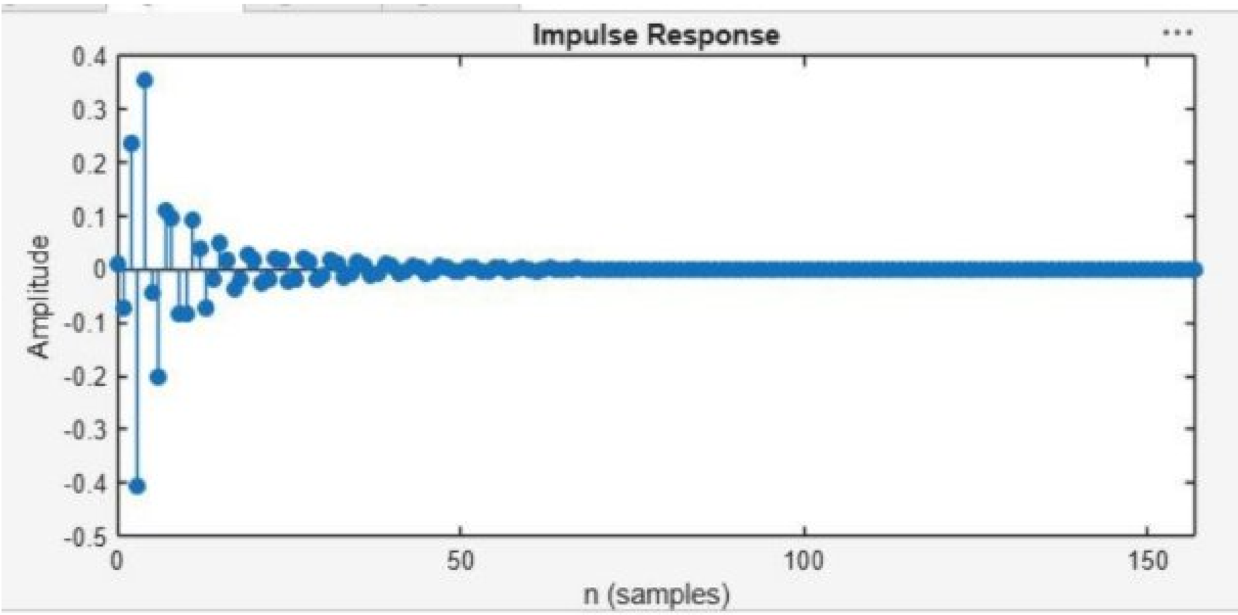
Filter Order = 6

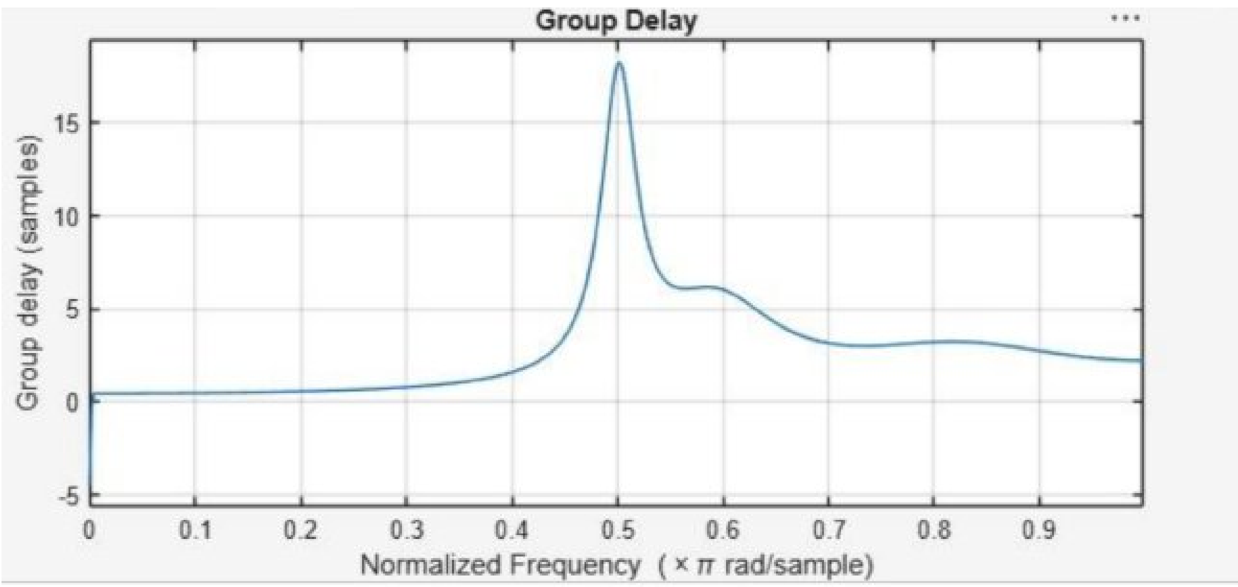
Cutoff Frequency = 0.5000

Objective 2:



Objective 3:


Impulse Response
Amplitude
0.4
0.3
0.2
0.1
0
-0.1
-0.2
-0.3
-0.4
-0.5
0
50
100
150
n (samples)


Group Delay
Group delay (samples)
15
10
5
0
-5
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
Normalized Frequency (× π rad/sample)

Objective 4: